

cpufreq (DVFS)

Introduction

The cpufreq subsystem allows for dynamic voltage and frequency scaling (DVFS), where a CPU's relative performance (and thus power consumption) can be tuned to the current workload.

This document outlines how to interact with the cpufreq subsystem from userspace to set a CPU's frequency as well as how to pin tasks to particular CPUs.

Feedback

How do we identify which CPUs are in the system?

Linux exposes this information through `/proc/cpuinfo`:

```
# cat /proc/cpuinfo
processor : 0
BogoMIPS : 100.00
Features : fp asimd evtstrm aes pmull sha1 sha2 crc32
CPU implementer : 0x41
CPU architecture: AArch64
CPU variant : 0x0
CPU part : 0xd03
CPU revision : 0

processor : 1
BogoMIPS : 100.00
Features : fp asimd evtstrm aes pmull sha1 sha2 crc32
CPU implementer : 0x41
```

```
CPU architecture: AArch64
CPU variant : 0x0
CPU part : 0xd07
CPU revision : 0

processor : 2
BogoMIPS : 100.00
Features : fp asimd evtstrm aes pmull sha1 sha2 crc32
CPU implementer : 0x41
CPU architecture: AArch64
CPU variant : 0x0
CPU part : 0xd07
CPU revision : 0

processor : 3
BogoMIPS : 100.00
Features : fp asimd evtstrm aes pmull sha1 sha2 crc32
CPU implementer : 0x41
CPU architecture: AArch64
CPU variant : 0x0
CPU part : 0xd03
CPU revision : 0

processor : 4
BogoMIPS : 100.00
Features : fp asimd evtstrm aes pmull sha1 sha2 crc32
CPU implementer : 0x41
CPU architecture: AArch64
CPU variant : 0x0
CPU part : 0xd03
CPU revision : 0

processor : 5
BogoMIPS : 100.00
Features : fp asimd evtstrm aes pmull sha1 sha2 crc32
CPU implementer : 0x41
CPU architecture: AArch64
CPU variant : 0x0
CPU part : 0xd03
CPU revision : 0

Hardware : ARM Juno development board (r0)
```

Here the `processor` number is the index the kernel uses for the CPU and corresponds to those used in `/sys/`, i.e. processor 1 in the above output

corresponds to `/sys/devices/system/cpu/cpu1/`.

The `CPU part` is the primary part number of the CPU as read from the `MIDR_EL1` system register. We can consult the Technical Reference Manual (TRM) of a processor to find the expected part number; for a Cortex-A53 this is `0xd03`, for a Cortex-A57 this is `0xd07`, and for a Cortex-A72 this is `0xd08`.

So in the example above CPUs 0, 3, 4, and 5 are Cortex-A53s because their part number is `0xd03`, while CPUs 1 and 2 are Cortex-A57 cores because their part number is `0xd07`.

Pinning processes to CPUs

In Linux each process has a CPU affinity describing which CPUs that process is allowed to run on in an SMP system; by default this is "all CPUs" but we can use the `taskset` command to choose the affinity.

Example 1 - Run `ls /lib` only on CPU 5 (Cortex_A53_4)

```
# taskset -c 5 ls /lib
```

Example 2 - Run `grep -i joe users.txt` allowed only on CPUs 1 and 2 (the two Cortex-A57 cores):

```
# taskset -c 1,2 grep -i joe users.txt
```

Example 3 - Run `ping 192.0.0.1` allowed on all CPUs:

```
# taskset -c 0-5 ping 192.0.0.1
```

Note: If the `taskset` command is not available, try using either the Android or OpenEmbedded LAMP filesystem.

Dynamic Voltage and Frequency Scaling (DVFS)

cpufreq

cpufreq allows for the clock speed of a CPU to be adjusted on-the-fly in order to adjust to current load demands; by reducing the clock speed while a processor is idle we can drastically reduce power consumption and heat generation.

To read the current operating frequency of a CPUm for example CPU 4:

```
# cat /sys/devices/system/cpu/cpu4/cpufreq/scaling_cur_freq  
1150000
```

You can obtain a list of available frequencies like so:

```
# cat /sys/devices/system/cpu/cpu4/cpufreq/scaling_available_frequencies  
600000 900000 1150000
```

And to select a frequency:

```
# echo 900000 > /sys/devices/system/cpu/cpu4/cpufreq/scaling_setspeed
```

Note: For Juno r1 boards the Cortex-A53 cores are limited to 650MHz only.

Scaling governor

The rules used by the kernel for dynamically adjusting the frequency are defined by the cpufreq scaling governor.

The governor defines the power characteristics of the system's CPUs which in turn affects CPU performance. Each governor has its own unique behaviour, purpose, and

use case suitability.

The available governors can be listed like so:

```
# cat /sys/devices/system/cpu/cpu4/cpufreq/scaling_available_governors
ondemand userspace interactive performance
```

To select a governor:

```
# echo userspace > /sys/devices/system/cpu/cpu4/cpufreq/scaling_governor
```

More information regarding cpufreq scaling governors can be found [here](#).

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Feedback

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